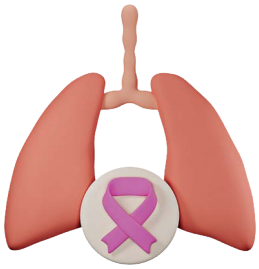
NEW HOPE FOR CHRONIC CONDITIONS

Precision medicine can transform the treatment of various diseases



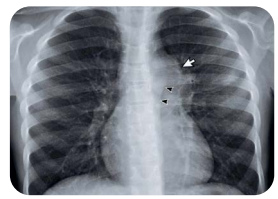
LUNG CANCER

Use: Tumour genome sequencing, targeted therapy

Success: Survival 2–3 times higher than chemotherapy

Cost: Test Rs 20,000–80,000; therapy Rs 1.5–3 lakh/month

Available at: AIIMS Delhi, Tata Memorial Hospital Mumbai, Narayana Health



DRUG-RESISTANT TUBERCULOSIS

Use: Genome testing detects drug resistance early

Success: Higher cure rates, faster correct treatment

Cost: Genomic test Rs 10,000–Rs 25,000

Available at: AIIMS Delhi, ICMR labs, MedGenome centres



RARE GENETIC DISEASES

Use: Gene therapy replaces defective gene

Success: Improvement in survival/ mobility

Cost: Therapy can cost several crores

Available at: AIIMS Delhi, Medanta Gurgaon, Apollo Hospitals



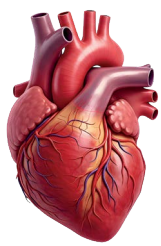
PARKINSON'S / ALZHEIMER'S

Use: Genetic testing identifies disease pathways

Success: Helps targeted therapy and clinical trials

Cost: Rs 20,000–Rs 50,000 for tests

Available at: NIMHANS Bengaluru, AIIMS Delhi, Apollo Hospitals



HEART DISEASE

Use: Genetic risk score predicts early heart disease

Success: Prevention can start years earlier

Cost: Test Rs 10,000–Rs 20,000

Available at: AIIMS Delhi, Medanta Gurgaon, Fortis Hospitals

While precision medicine began in cancer care, its use is now expanding, especially in rare genetic disorders, cardiology and neurology. Indian hospitals are using genomic testing panels (tests that analyse multiple genes linked to cancer) for treatment selection and pharmacogenomic tests to identify medicines a patient is more likely to respond to. Gene-based treatments for conditions like spinal muscular atrophy and inherited retinal disorders are becoming available through specialised centres.

The developments can't come soon enough. India accounts for 7–8 per cent of global cancer cases, nearly 27 per cent of the world's tuberculosis burden and has an estimated 70 million people living with rare diseases, many of them genetic.

The Indian precision medicine market is expected to grow rapidly as genome sequencing becomes cheaper and more hospitals use genetic testing before prescribing treatment. Market research firm IMARC put the size of the market at $2.67 billion (Rs 24,898 crore) in 2025 and estimates annual growth at around 11–12 per cent.

THE NEW CANCER PLAYBOOK

Cancer remains the most fertile ground for precision medicine. In many cancers where decisions are based on tumour location and extent of malignancy, doctors can now identify specific mutations driving the disease.

Rohit Menon, a 46-year-old chartered accountant from Kochi, was diagnosed with advanced non-small cell lung cancer. Instead of starting chemotherapy, his doctors performed tumour genome sequencing, which revealed an EGFR gene mutation. He took targeted therapy; within months, his tumour reduced and his disease stabilised. This is because targeted drugs block the mutation driving the cancer cells, rather than attacking all rapidly dividing cells the way chemotherapy does. For patients, this means fewer side-effects, longer disease control and a better quality of life.

A 2026 study in the *Journal of Clinical Oncology* reported that, in certain advanced cancers, patients whose treatment was guided by genome sequencing and matched targeted therapy had progression-free survival rates 30–40 per cent higher than those treated with conventional chemotherapy alone. Some cancers may be detected before symptoms appear. "Liquid biopsy tests can detect fragments of tumour DNA in blood, sometimes years before any symptoms, which could allow cancer to be treated earlier," says Dr Maurya.

However, much depends on tumour biology and disease stage. "Tumour genome sequencing is